LS-CAT Management Board Meeting

Agenda

Call In #: 1-888-850-4523
Passcode: 190580

http://conf1.ls-cat.org
http://conf2.ls-cat.org

All times CST

Location: Building 436C Conference Room

Noon	Buffet lunch and LS-CAT tour
2:00	Approval of minutes and agenda
2:10	Dennis Mills – Upgrade plans: what the CATS should be doing to prepare
2:40	Keith Brister – LS-CAT status
3:00	David Smith – Robot upgrades
3:10	Elena Kondrashkina – Beam stability and beamline controls
3:30	Joseph Brunzelle – User science
3:50	Spencer Anderson – Eiger data collection parameters
4:10	Keith Brister – Usage statistics
4:30	Closed session
5:00	Meeting adjournment
5:30	Dinner at Argonne Guest House

What's next?

It's not too early to start detailed plans for LS-CAT after the APS upgrade.

1) What is the new science that the upgraded APS will enable?
2) How do we go about building a world class facility for the mid-2020's?
3) What direction should each of our stations take?
   a) Simple plan: upgrade robots, diffractometers, detectors. Refurbish or replace optics.
   b) Complex plan: above plus new crystal delivery techniques.
   c) Doing nothing is not feasible as our equipment is becoming obsolete. Some items already cannot be easily repaired or replaced.
4) Where are the big pots of money? This will not come cheap.

Post APS Upgrade Challenges

1) The DOE may mandate 40% of our time for general users instead of the current 25%.
   a) There are currently not enough general users to meet this requirement.
   b) The efficiencies required to allow our members to collect all their own data as well as an extra “shares” worth of time for the GUP would require a larger staff (and budget).

Post APS Upgrade Challenges

2) ID-19 and ID-20 are planned to become long beamlines.
   a) This will put a beam pipe through our lab
   b) and separate our beam lines from our offices.
   c) Following slides are from Dean Haeffner (APS/APS-U)

Selected Beamline Proposals

Name	Title	Technique
CHEX	Coherent High-Energy X-ray Sector for In Situ Science	*In situ*, surface high-energy coherent scattering
Polar	Polarization modulation spectroscopy	Magnetic spectroscopy
HEXM	A High-Energy X-ray Microscope	High-energy microscopies & CDI
SAXPCS	Development of a Small-Angle X-ray Photon Correlation Spectroscopy Beamline for Studying Dynamics in Soft Matter	Small-angle XPCS
WAXPCS	Wide-Angle X-Ray Photon Correlation Spectroscopy and Time-Resolved Coherent X-Ray Scattering Beamline	Wide-angle XPCS
Ptycho	PtychoProbe	Ultimate resolution, forward scattering ptychography/spectromicroscopy
InSitu	In Situ Nanoprobe Beamline	*In situ*, forward scattering ptychography/spectromicroscopy Long working distances
CSSI	Coherent Surface Scattering Imaging Beamline for Unraveling Mesoscopic Spatial-Temporal Correlations	Coherent GISAXS, XPCS
Atomic	Atomic – A beamline for extremely high resolution coherent imaging of atomistic structures	Diffraction microscopy & CDI Bragg CDI
3DNano	3D Micro & Nano Diffraction	Upgrade of current 34-ID

Roadmap Considerations – Long Beamlines

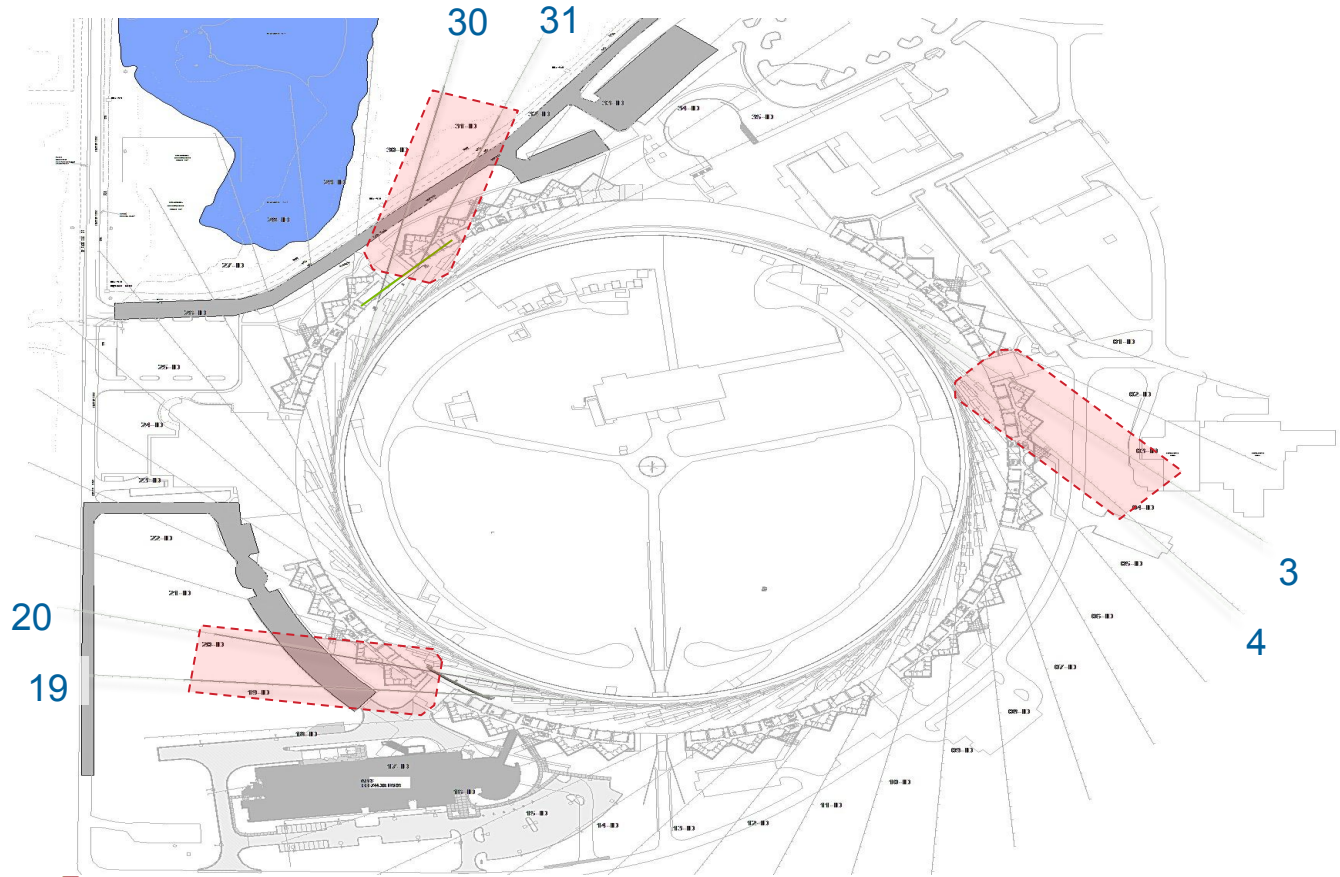
- Standard APS beamlines limited to ~75 m on the Experiment Hall floor
- Beamlines that extend beyond the Experiment Hall floor are “Long Beamlines”
  - None currently at the APS
  - Common feature at ESRF, NSLS-II, Spring-8, Diamond, etc.
- APS-U as an MIE Project is limited to $10M civil construction
- Need two long beamlines (HEXM, InSitu)
    - InSitu ideally be around 250 m from source point
    - HEXM more flexible
    - Would like to site together
- Factors
  - Impact on existing beamlines
  - Impact on LOMs
  - Environmental issues
    - Wetlands, etc.
  - Impact on non-APS infrastructure
    - Roads
    - Utility lines, sewers, etc.
  - Topography

Long Beamlines Process

- Brought in John Sidarous, who recently retired, back as an STA to help
- Engaged StudioGC, a Chicago based architectural firm to study scenarios
- Met with representatives of the various Argonne interests
  - Environment, utilities, plant and facilities, etc.

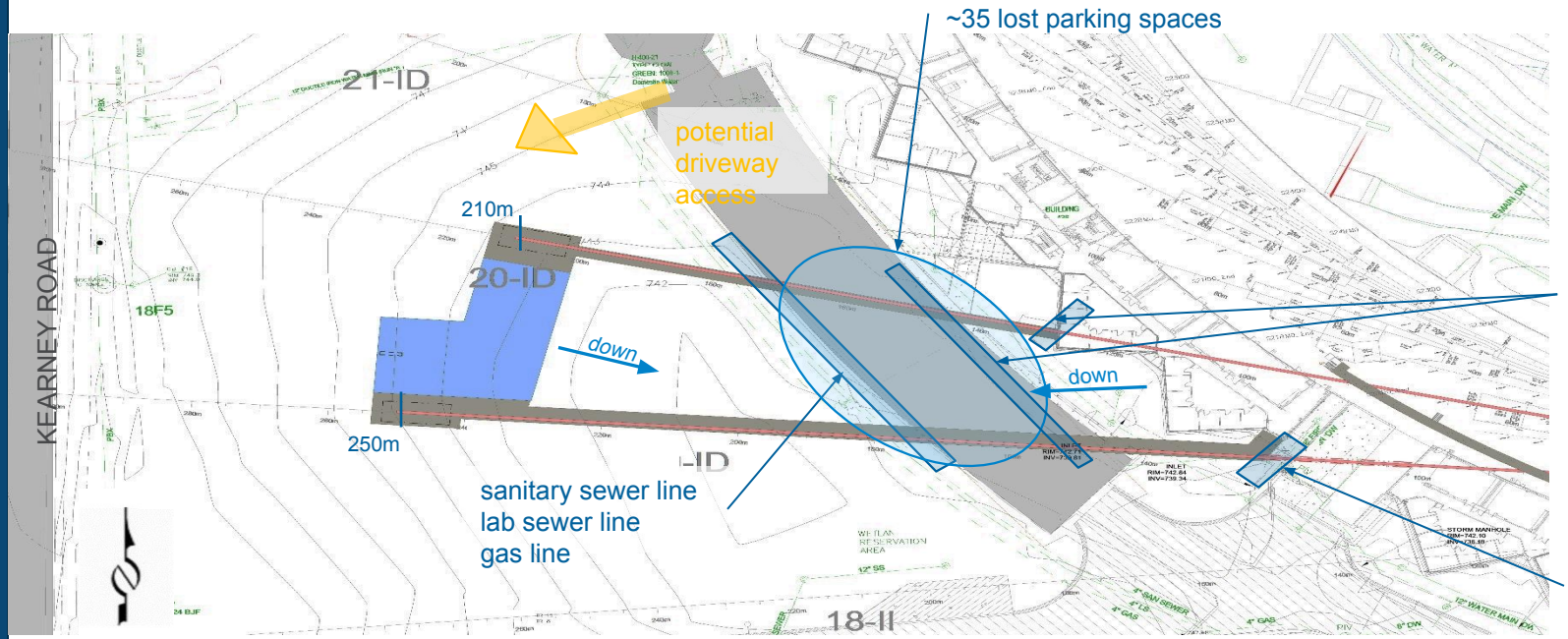
- Looked at all APS sectors as potential sites
- Developed a set of options based on three areas that were the most viable

Long Beamlines Overview



Argonne NATIONAL LABORATORY

Long Beamlines – Sectors 19 & 20 Scenario

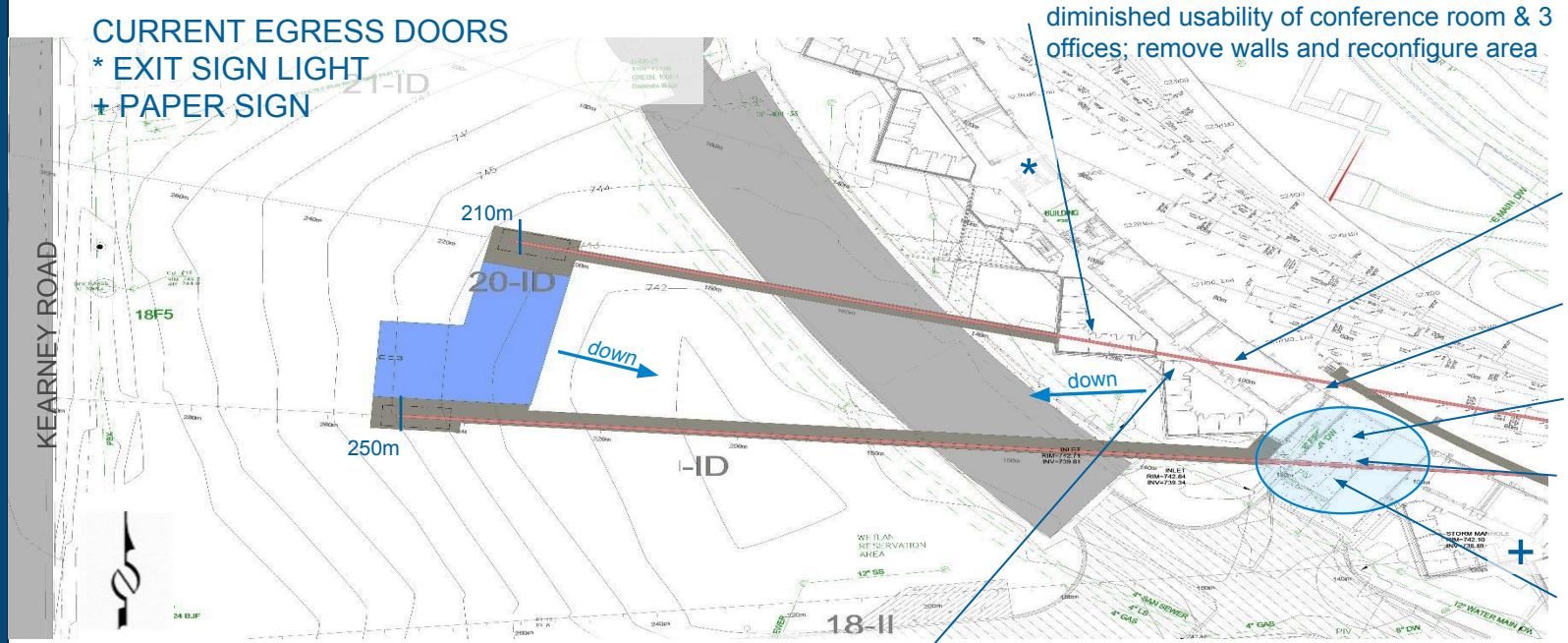


SITE CONSIDERATIONS EVALUATION												
BEAM LINES	TOPO	SOILS & FOUND.	ROADWAY DISRUPTION	LOST PARKING	UTILITIES *(scale 1-20)*	NATURAL RESOURCES	STORM WATER DETENTION	RING ACCES.	LOM RECONFIG	LOM SUPPORT	BUILDING 400 SUPPORT	OVERALL SCORE
19+20	8 **	7	10	5	12	10	6	5 *(elec panel)*	6	8	6	**83**

** *Approximately 2' drop and/or gain in elevation*

Scale: 1-10, 1 being the least desirable

Long Beamlines – Sectors 19 & 20 Scenario



SITE CONSIDERATIONS EVALUATION

BEAM LINES	TOPO	SOILS & FOUND.	ROADWAY DISRUPTION	LOST PARKING	UTILITIES *(scale 1-20)*	NATURAL RESOURCES	STORM WATER DETENTION	RING ACCES.	LOM RECONFIG	LOM SUPPORT	BUILDING 400 SUPPORT	OVERALL SCORE
19+20	8 **	7	10	5	12	10	6	5 *(elec panel)*	6	8	6	**83**

** *Approximately 2' drop and/or gain in elevation*

Scale: 1-10, 1 being the least desirable

Long Beamlines – Site Summaries

SITE CONSIDERATIONS EVALUATION													
BEAM LINES	**TOPO**	**SOILS & FOUND.**	**ROADWAY DISRUPTION**	**LOST PARKING**	**UTILITIES** *(scale 1-20)*	**NATURAL RES.**	**STORM WATER DETENTION**	**RING ACCESS**	**LOM RECONF.**	**LOM SUPPORT**	**BUILDING 400 SUPPORT**	**SCORE**	**OPTIONS**
19+20	8	7	10	5	12	10	6	5	6	8	6	**83**	**F / G**
30+31	2	5	3	10	6	5	6	6	5	8	4	**60**	**F / G**
20+21	7	7	10	5	12	10	6	5	2	1	7	**70**	**A, B, C, D, E, F AND G**
31+32	5	5	3	8	4	5	6	6	4	1	7	**54**	**A, B, C, D, E, F AND G**
3+4	1	5	1	5	10	8	4	4	2	1	4	**45**	**A, B, C, D, E, F AND G**

Scale: 1-10, 1 being the least desirable

Long Beamlines – Conventional Facilities Costs

- Cost Estimations
  - From John Sidarous
  - Bottom line

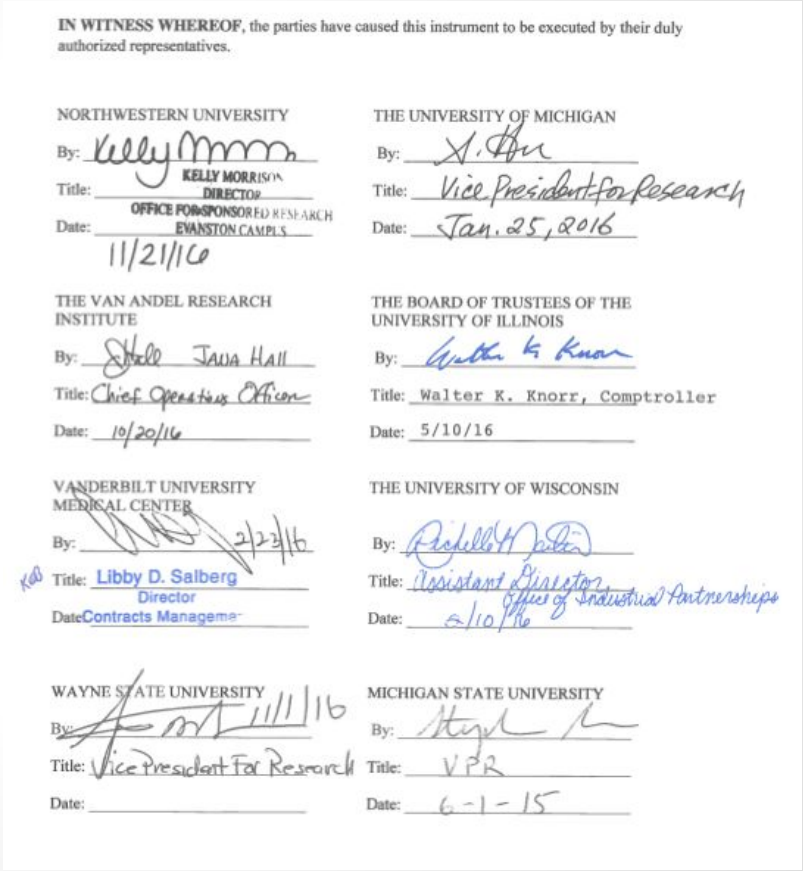
Scenario	Base Cost ($M)	Loaded Cost ($M)	With 35% Contingency ($M)
19 & 20	6.987	8.290	11.191
30 & 31	7.752	9.197	12.416

  - Sector 3 & 4 scenario not fully costed due to uncertainties in utilities, cost of impact on Building 314, but is thought to be considerably more than Sector 30 & 31 scenario

LS-CAT Status

LS-CAT Extension Signed!

The fourth amendment to the LS-CAT agreement extends the end date of the contract to February 28, 2022.



IN WITNESS WHEREOF, the parties have caused this instrument to be executed by their duly authorized representatives.

NORTHWESTERN UNIVERSITY
By: Kelly Morrison
Title: KELLY MORRISON DIRECTOR OFFICE FOR SPONSORED RESEARCH EVANSTON CAMPUS
Date: 11/21/16

THE UNIVERSITY OF MICHIGAN
By: [signature]
Title: Vice President for Research
Date: Jan. 25, 2016

THE VAN ANDEL RESEARCH INSTITUTE
By: Jana Hall
Title: Chief Operating Officer
Date: 10/20/16

THE BOARD OF TRUSTEES OF THE UNIVERSITY OF ILLINOIS
By: [signature]
Title: Walter K. Knorr, Comptroller
Date: 5/10/16

VANDERBILT UNIVERSITY MEDICAL CENTER
By: [signature] 2/23/16
Title: Libby D. Salberg Director
Date: Contracts Manageme

THE UNIVERSITY OF WISCONSIN
By: [signature]
Title: Assistant Director Office of Industrial Partnerships
Date: 2/10/16

WAYNE STATE UNIVERSITY
By: [signature] 11/1/16
Title: Vice President For Research
Date:

MICHIGAN STATE UNIVERSITY
By: [signature]
Title: VPR
Date: 6-1-15

Bionanoprobe

The bionanoprobe has been successfully moved from 21-ID-D to 19-ID-B.

At this point we are no longer involved in the day-to-day operations of the instrument.

This frees up the schedule but also makes 21-ID-D (and its Eiger detector) available to MX general users.



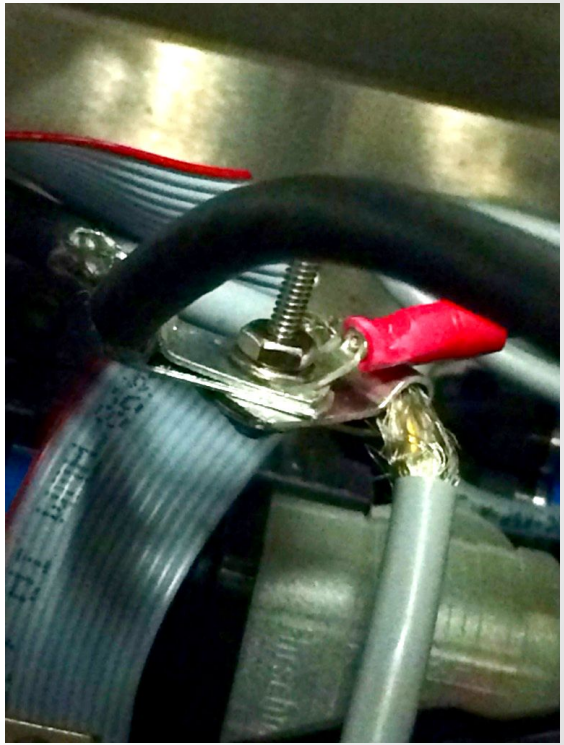
Amusing Failure 1: LN2 Fill Pipe





Amusing Failure 2: Omega on 21-ID-D

For reasons we do not fully understand the manufacturer had stripped the insulation from the omega axis drive cable and omega axis encoder cable and then clamped the two together in the fashion shown in the photo. Starting this Fall this design error has led to the omega axis failing to move from time to time. Currently our fix is to slightly wiggle the bundle of wires.

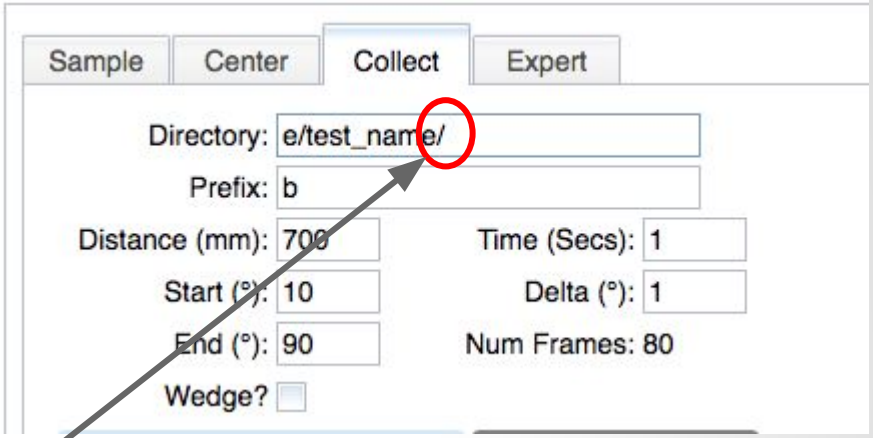


Amusing Failure 3: Much ado about nothing

A bug developed that caused the remote interface to start requesting that the server create millions of blank frame definitions.

This caused data collection on the side stations to slow to a crawl as the database struggled to keep up with the task of keep track of all these useless frames. It only happened occasionally and only for certain users.

Fix was odd but simple, to remove the trailing slash on directory names whenever one was used.



Amusing Failure 4: Jitter Bug

The cameras had been plagued by some jitter prior to this run. It was annoying but easy to fix and the fix would work for hours at a time.

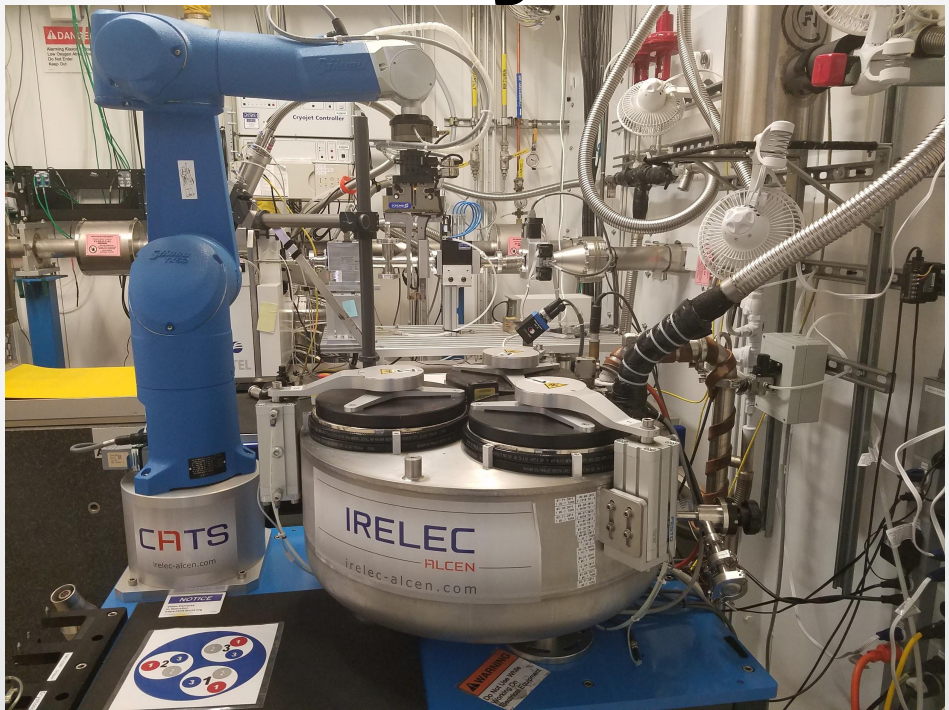
This run the jitter made the stations unusable and the fix would work for only minutes, not hours.

The source seems to be a sensitivity to network congestion in the camera code coupled with an increase in network congestion. One cause was a misconfigured ethernet port at LS-CAT.

A temporary fix for the sensitivity itself is in place now and a better fix is planned for coming down period.



Robot Upgrade Plans
January 2017

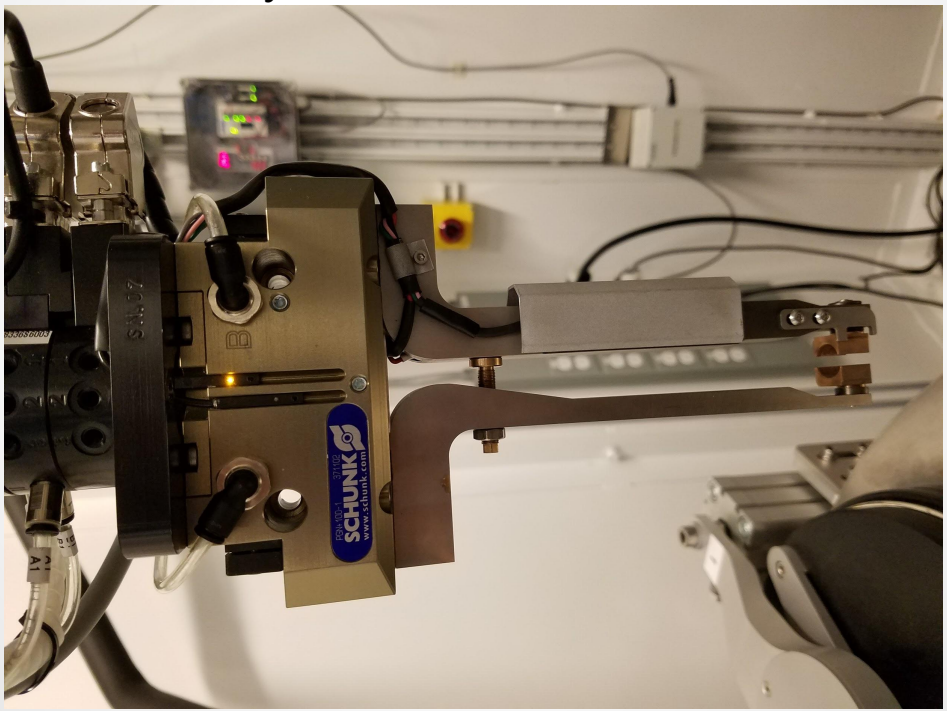


Current State of Robots

- Cycle times on the robots are ~ 92 ± 3 s per station for dismount/mount cycle
- Reliability > 99.5%
- Retraining to the goiniometer down to ~ 1 or 2 times per run per station
- Crashes in robot dewar or goiniometer down to less than once per week
- Grippers maintained each run

Jan 2017 Upgrade plans

- New motors on EMBL grippers and optimized paths will decrease cycle time of dismount/mount by ~20s *



*Irelec claim

- Improved logging of events (software)
- Upgraded positional tracking of the robot should lead to faster recovery times after robot halting events (crash, abort, panic)
- Log file will now contain more detailed information related to why a path has stalled which will lead to much faster recovery from unusual path breaking events.
    - Limit switch issues
    - Puck removal issues
    - Trajectory issues
- Gain the ability to speed up and slow the robot remotely (from the user GUI)
- Prevents certain commands during running of a path (such as clear memory)
- Logs the rotation time of the gripper so that we can do preventative maintenance prior to a rotation failure which requires swapping the grippers during user operations
- Implement magnet timing on D, E, and F which will dramatically decrease the number of dropped caps on D

Beam stability and beamline controls

Elena Kondrashkina

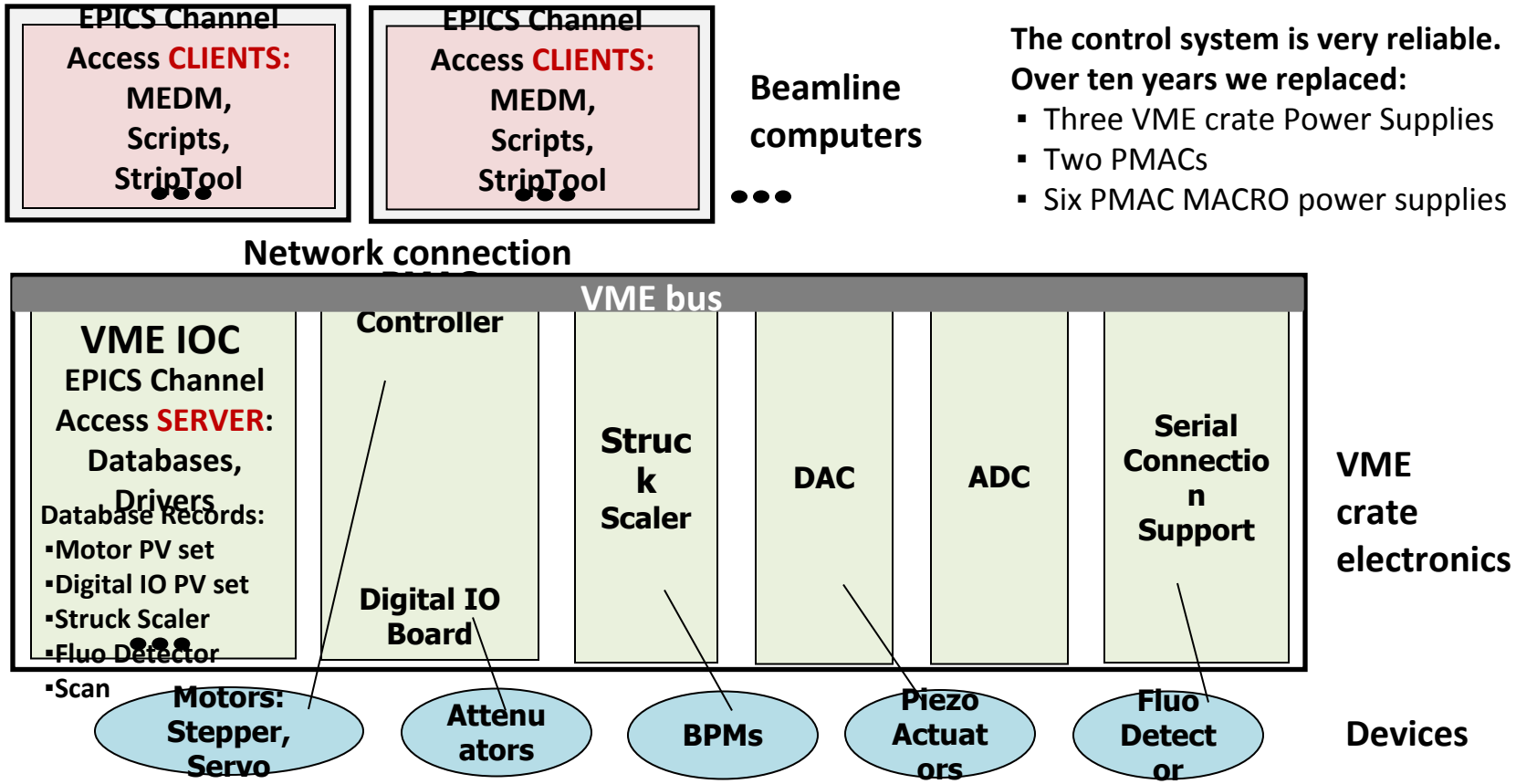
Responsibilities at LS-CAT

- Motion controls for beamline components (EPICS).
- Beam monitors, attenuators, Fluorescence Detector control.
- Scripts for beam energy tuning , feedback and scans.
- Beam focusing.
- Beamline alignment.
- User support.

Previous Experience

- PhD from the Institute of Crystallography, Moscow, Russia.
- Maintaining laboratory X-ray setups at Humboldt University, Germany and Rigaku Americas, USA.
- Beamline Scientist at APS, Sector 18. Experiments at DESY, Germany and ESRF, France.
- X-ray techniques: surface diffraction, small-angle scattering, crystallography for material science and

EPICS - Experimental Physics and Industrial Control System



The control system is very reliable.
Over ten years we replaced:

- Three VME crate Power Supplies
- Two PMACs
- Six PMAC MACRO power supplies

***MD2, video-cameras, Eiger&MAR detectors are controlled by K.Brister's software

EPICS controls location on the roof of LS-CAT hutches

C1-2

D1 electronics cabinet



Rear view

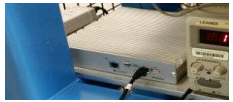


Front view

VME PMAC 2006, currently discontinued

Libera BPM





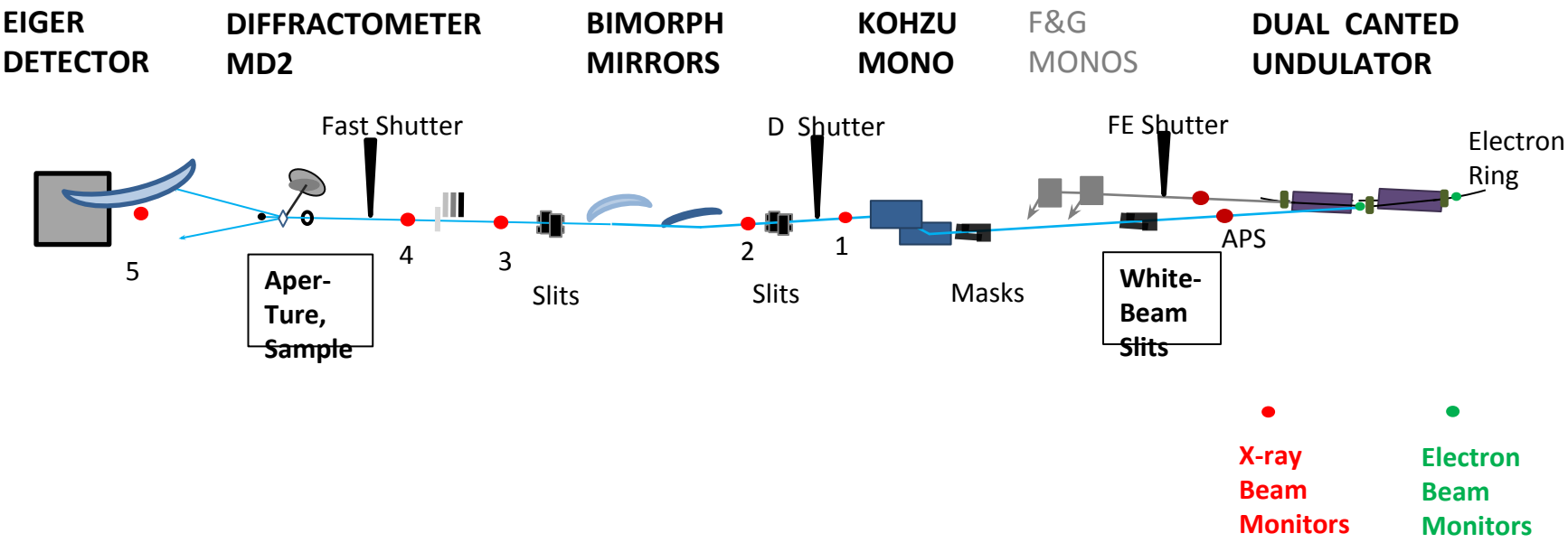


Replacement:
Power PMAC with embedded EPICS soft IOC

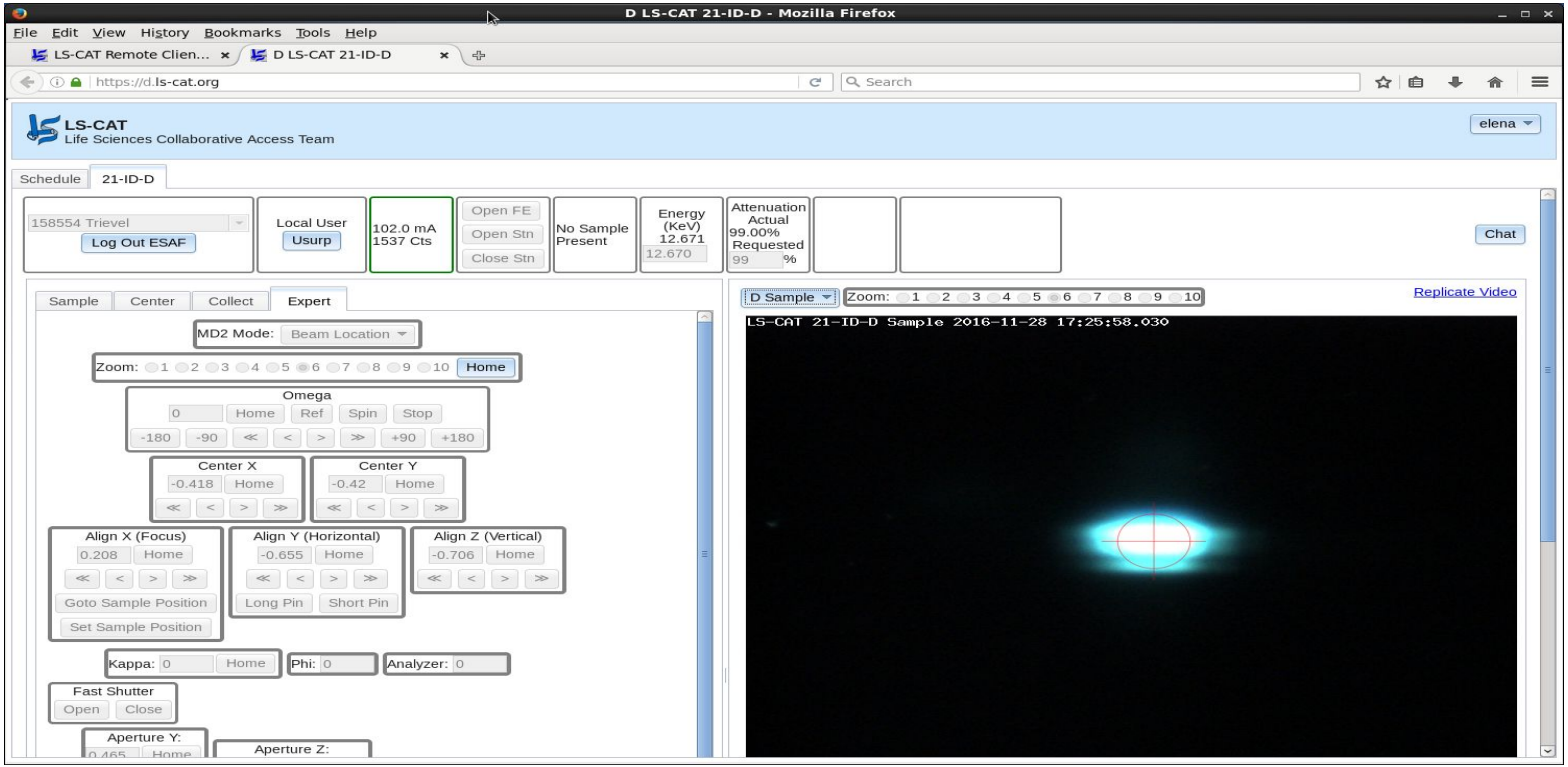
Other embedded EPICS soft-IOCs already running on D beamline:

- Libera BPM controls
- Mirror-bending control
- IOC for SNL scripts: Change-Energy and Fdbk

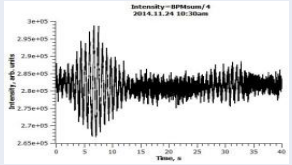
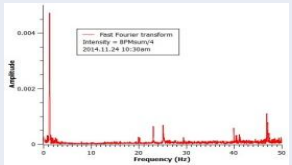
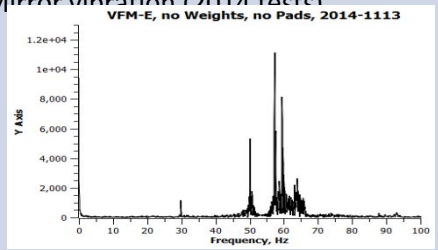
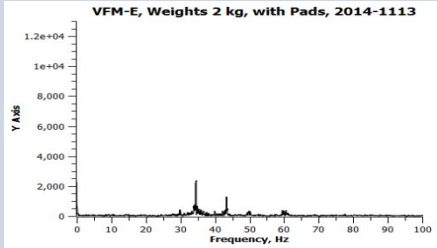
21ID-D Beamline Layout



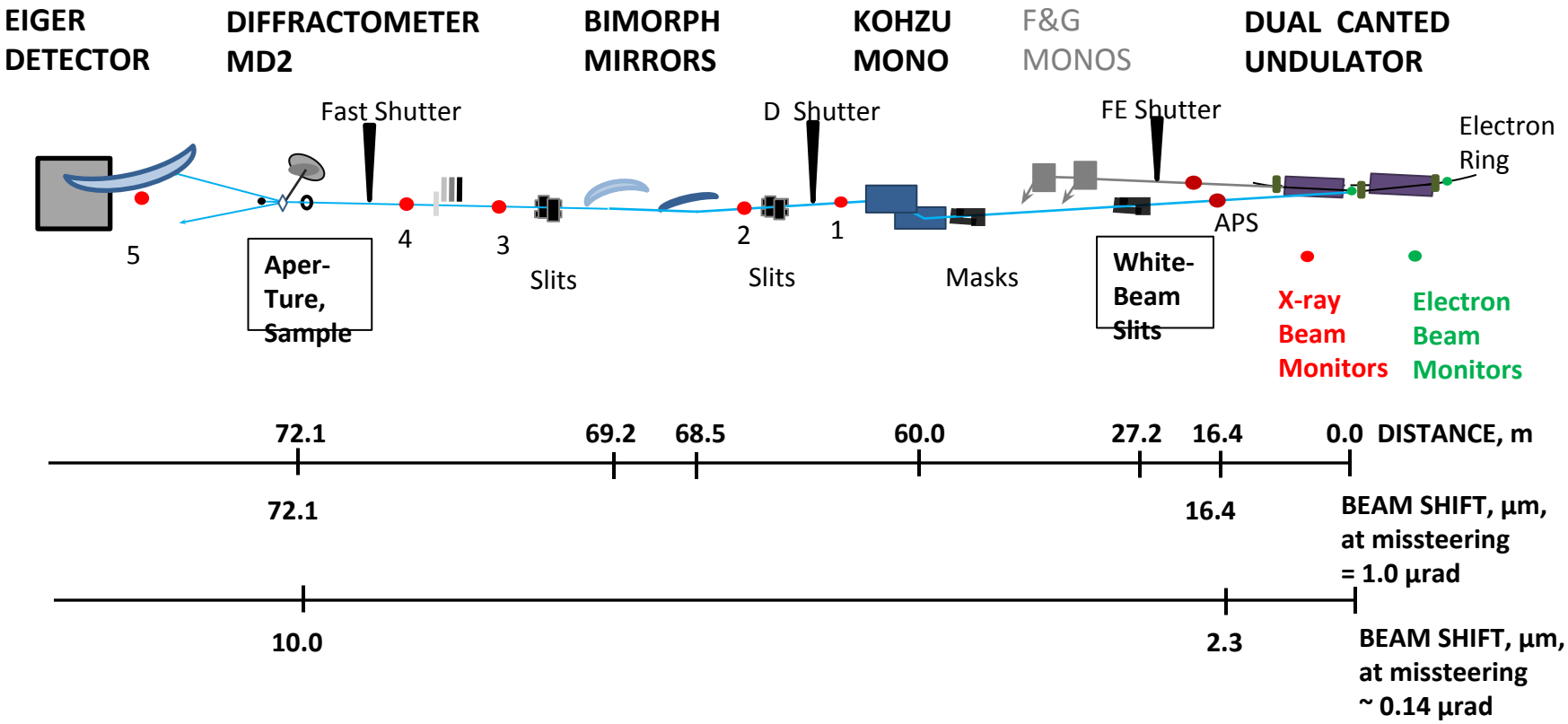
Best Beam on D beamline



Beam stability

Instability	Time scale	Cause (examples)	Solution
VIBRATIONS	~1 Hz	Faulty APS X-Ray BPM (2012 event) 	Remove and repair the APS BPM
	~1 Hz	KOHZU Mono problem (2014 event)   	Open the vacuum tank and reassemble Mono. Adjust pump frequency, LN2 plumbing, crystals gluing.
	30-65 Hz	Mirror vibration (2014 tests)  	Weights onto and pads under the mirror support

21ID-D Beamline Layout



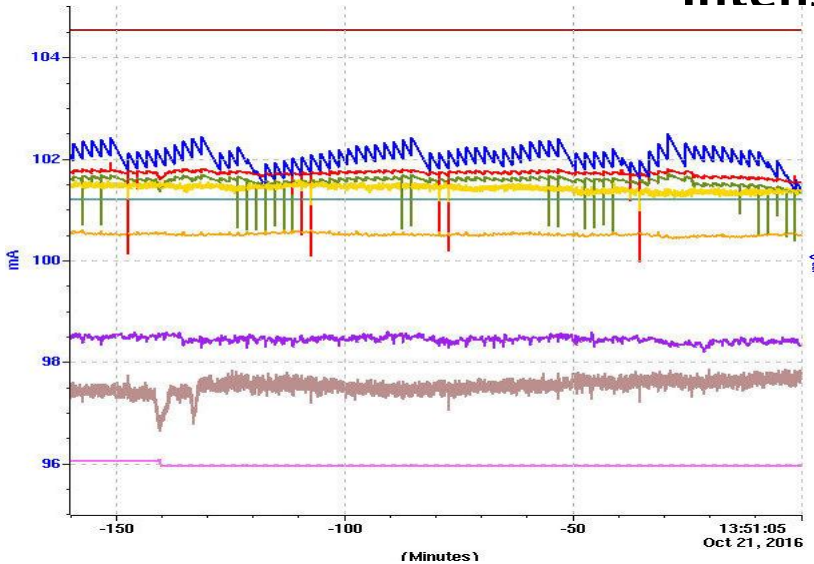
Beam stability

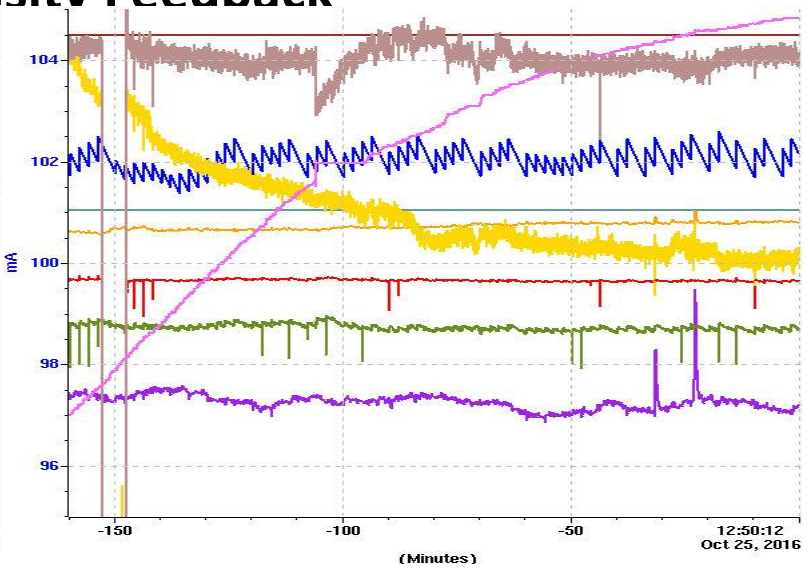
Instability	Time scale: Event	Cause	Solution
SHIFT	Beam steered	Missteering by APS or BL staff	Check beam position with the White-Beam slits and steer the beam
		Cross-dependence of the DS and US Undulators at steering the inboard beam (2012)	APS can slightly improve Upgrade of the Undulators
	Energy changed	Undulator design and operation	APS can slightly improve Upgrade of the Undulators
		Monochromator: beam constant-exit is not perfect	Upgrade of the Mono Improve alignment
	Beam focused Mirror Lane changed	Mirrors installation and alignment	Improve alignment

Beam stability, 2016 development

Instability	Time scale	Cause	Solution
DRIFT	Minutes-Hours	Mulitiple, each case requires investigation First suspects: APS, KOHZU MONO	Intensity Feedback Positional Feedback Beam Auto-Alignment

Intensity Feedback



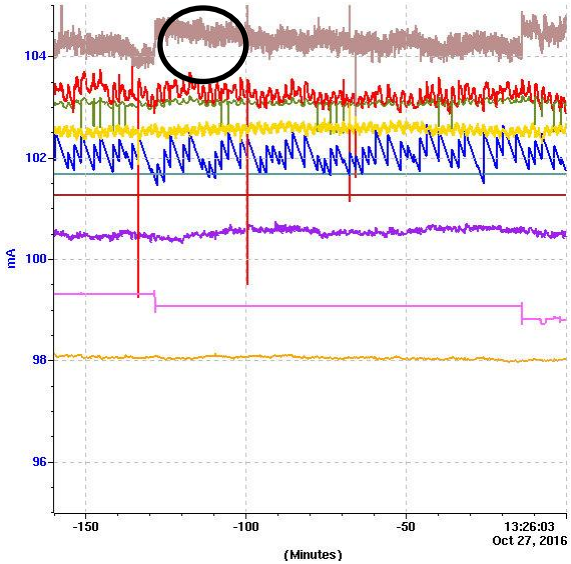


Piezo V
Y-position
BPM2
Ring Current
X-position
BPM2
Intensity
BPM 1

En=12.663 keV
Mono piezo is nearly idle.

En=13.57 keV, 3d harmonics, heat load ~10 times higher than at 12.6keV
Mono piezo voltage has changed by ~1 V

Sensitivity of LS-CAT BPMs to Beam Position



Y-position BPM2

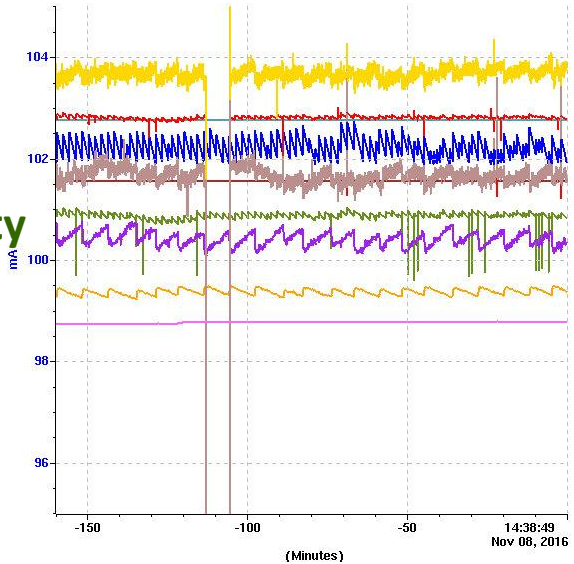
X-position BPM2

Ring Current

Intensity

BPM 1

Piezo V

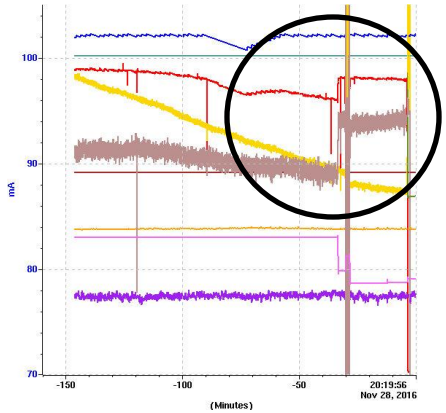


Beam adjusted by user by ~10 µm vertically at the sample location.

APS operation mode:
Reduced Horizontal Beam Lattice (RHB)

APS mode of operation:
Hybrid Fill RHB (Singlet)

Beam Stability Examples, Nov.- Dec. 2016

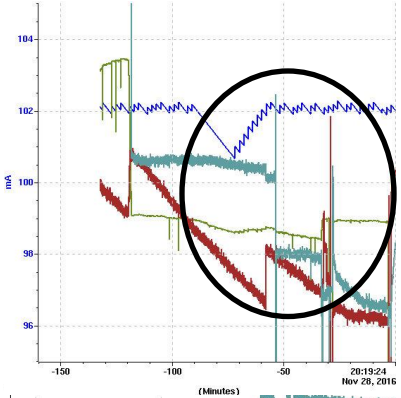


Ring Current

Y-position
BPM2

X-position
BPM2

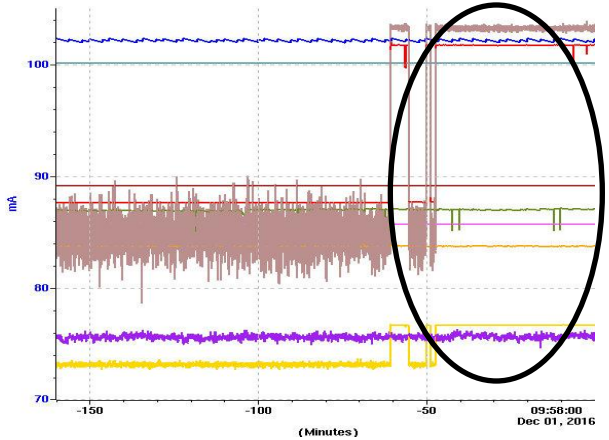
Piezo V



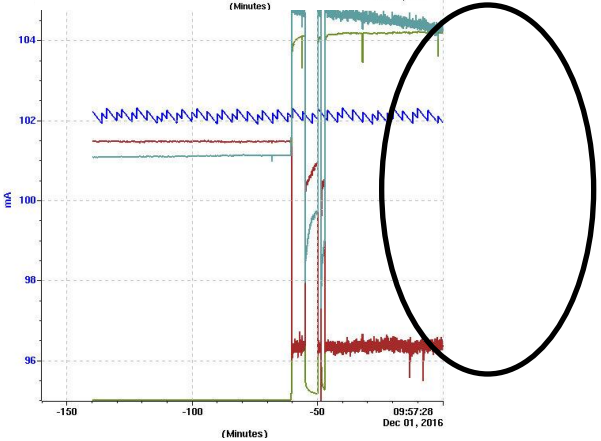
Beam unstable horizontally, APS was a cause

X-position
BPM3

Y-position
BPM3



Intensity
BPM1



Beam stable horizontally

Beam Position Feedback

What is done

- All five beam monitors are installed at the beamline and in a working condition.
- User Hutch BPMs provide sensitivity to the beam position better than 10 µm.
- Feedback EPICS-SNL (State Notation Language) script is easily extendable for Positional Feedback.
- Beam stability qualitative statistics is collected.

Challenges

- Beam-stability problems happen unpredictably and then Positional Feedback can tune the beam further away.
- Beam steering and tuning energy change scale of absolute readings from BPMs.
- User control of Positional Feedback is hardly possible, staff member will be needed.

Solution

- Other PX beamlines on the ring prefer using auto-alignment of the beam

Conclusions

- EPICS turned to be a very reliable system over the ten-year period.
- Upgrade of beamline controls has already started.
- Beam Auto-alignment button is the first priority.
- Beam monitoring and collaborating with APS Diagnostic Group and Accelerator Operations Physics Group, ASD needs to be continued.

Acknowledgements

K.Brister for guidance and discussions at all steps of the work

M. Bolbat, J. Ackley and J. VonOsinski for collaboration on installations and wiring of the beamline controls.

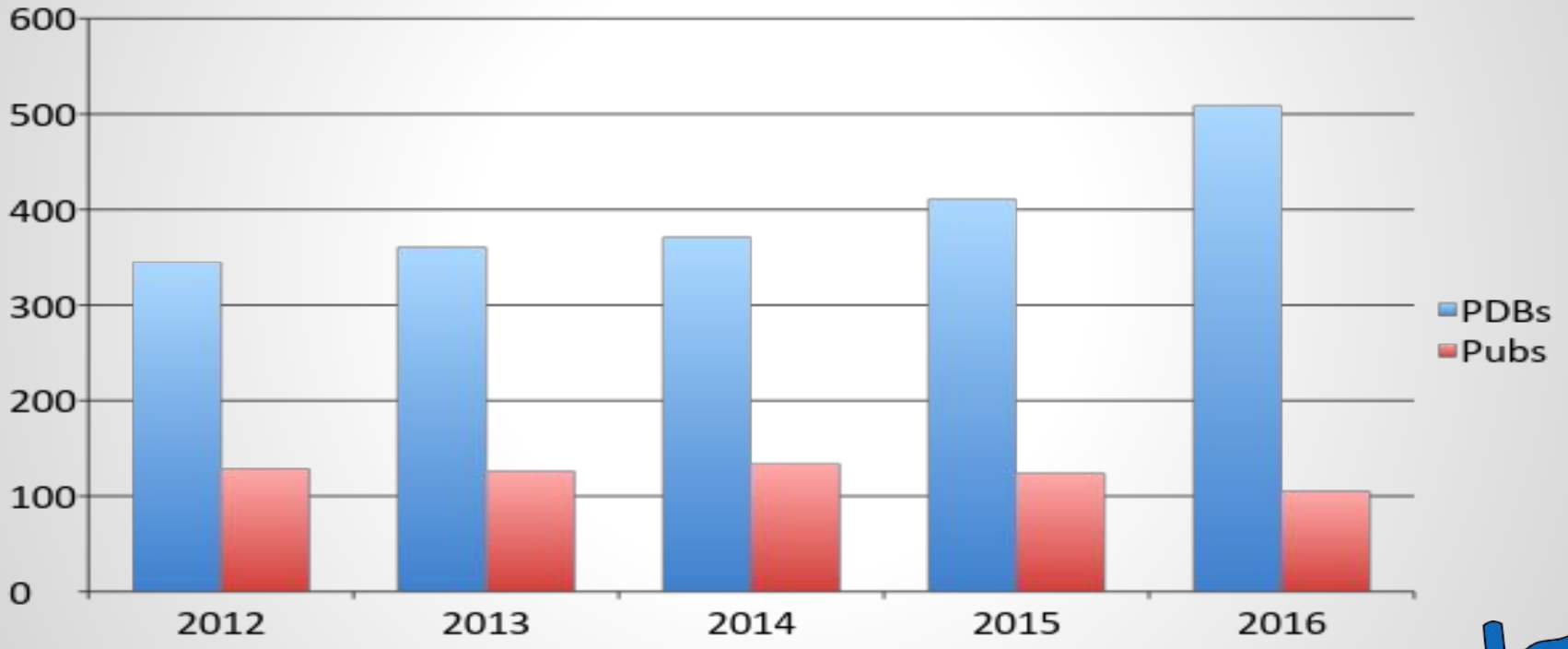
APS, GM-CA/ O. Makarov and S. Stepanov for EPICS development and help

Science at LS-CAT

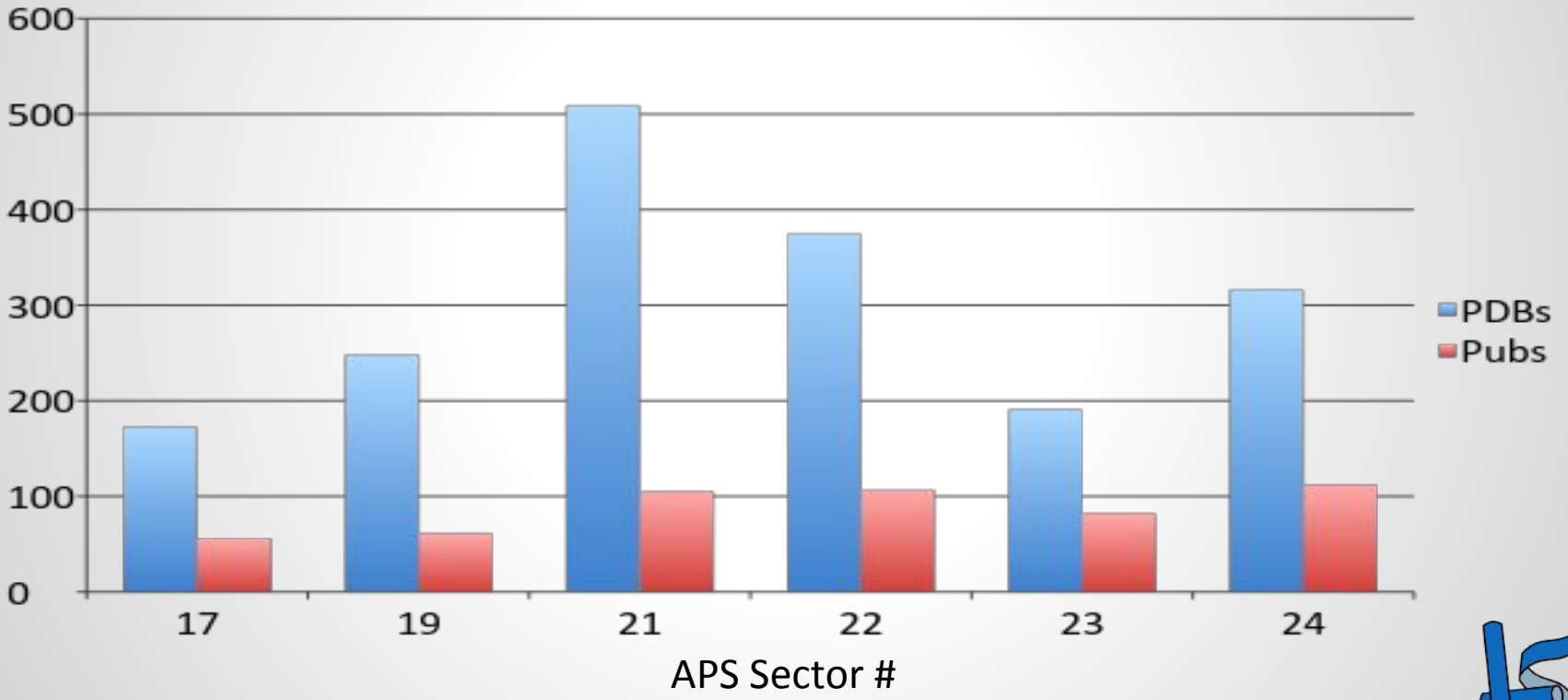
Joseph S Brunzelle



LS-CAT 5 Year Productivity



2016 Beam line Productivity Metrics



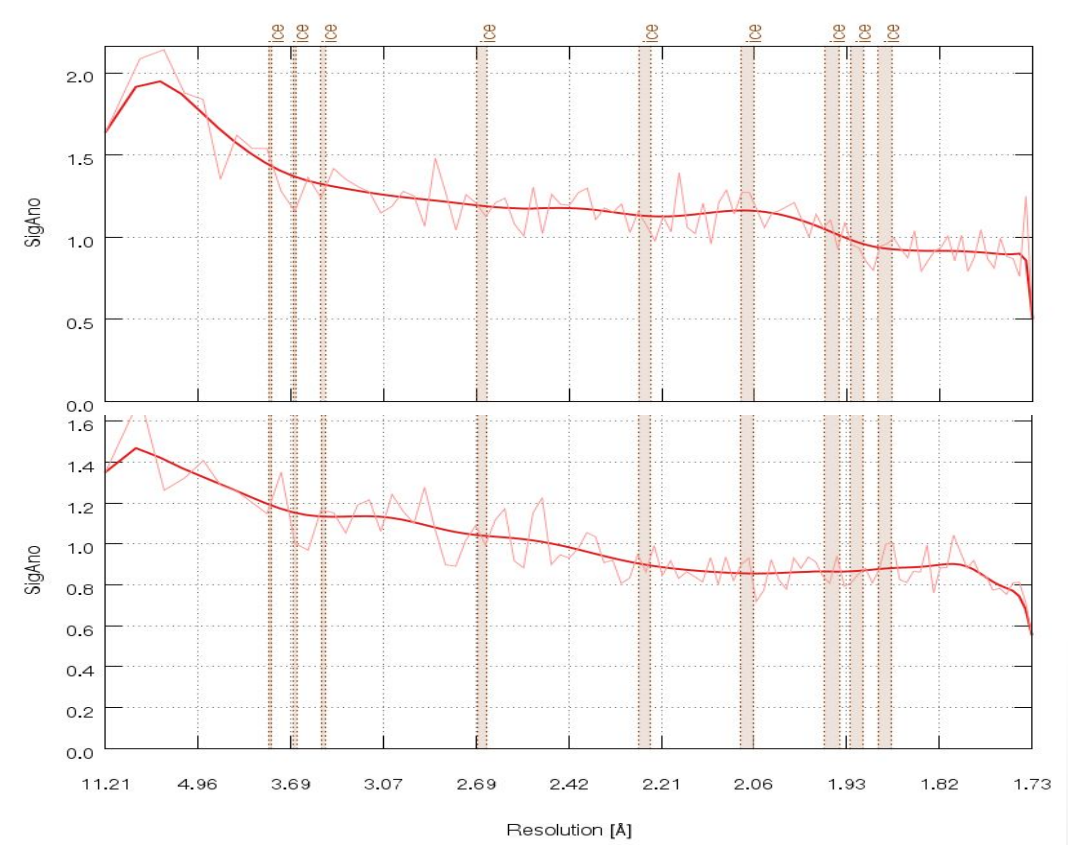
What do we publish?



Sulfur Phasing of Insulin

	001 S	006 S
Energy/Wavelength	8keV/ 1.55A	
% Attenuation	99	65
Frame Time (sec)	0.25	0.00625
Frame Width (deg)	0.5	0.5
Resolution (A)	1.725	1.726
Rmerge (%)	7.5	7.6
<I>/sigI	38.5	36.3
Anom Multiplicity	16.3	13.5
|DANO|/sd(DANO)	1.163	0.978
Total Time Exposed (secs)	180	4.5
#Sites	6	7
FOM	0.41	0.37
Overall model-map CC	0.872	0.839
Res Built	47	43

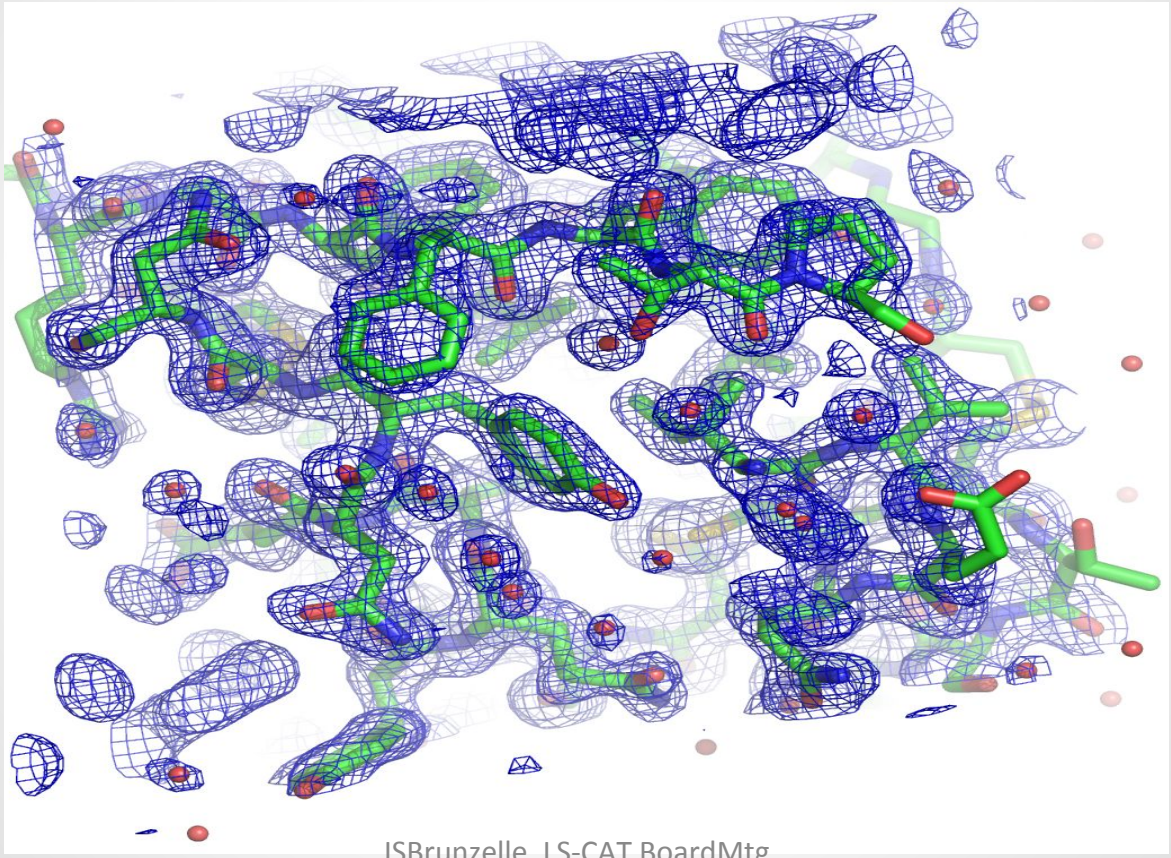
Speed and Anomalous Signal



180 sec dataset

4.5 sec dataset

Initial Model via Phenix



Training

- Location
  - Onsite at LS-CAT
  - At remote site
- Content
  - Sample mounting
  - Choosing beam line parameters
  - Fine and ultra-fine phi-slicing
  - Initial and final data analysis

Data Collection with the Eiger 9M

Spencer Anderson

LS-CAT Board Meeting

12/8/16

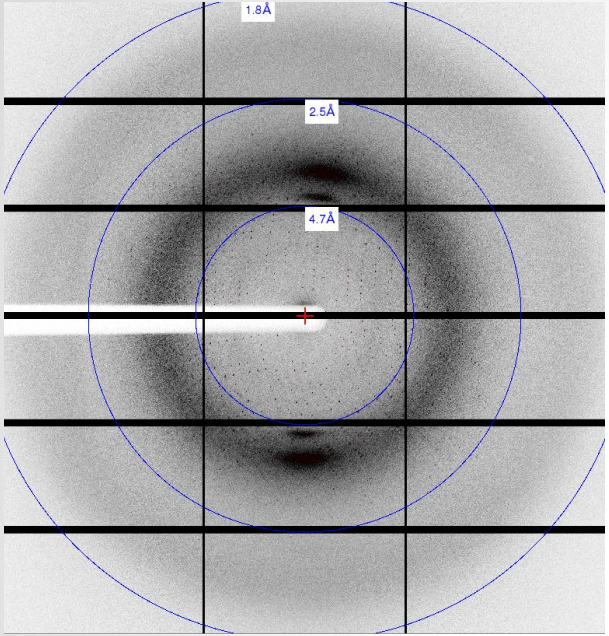
- In operation for almost a year
- How are people using it?
- How can we adapt to facilitate user needs?

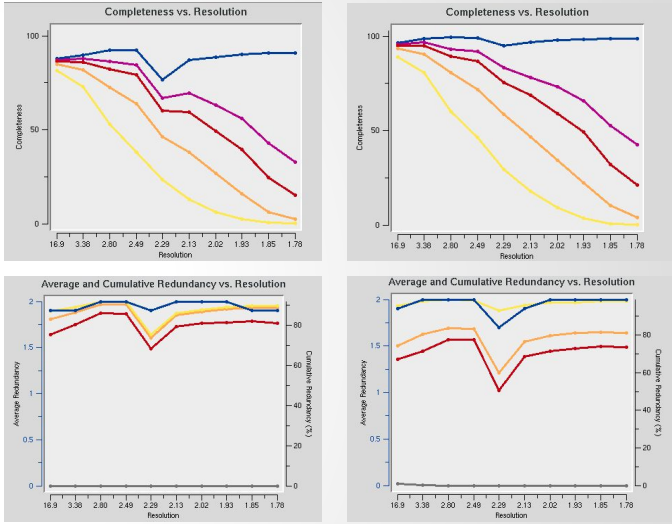
Replaced MX225 with Eiger 9M

- Similar area
  - Eiger 233 x 245mm vs 225 x 225mm
- Similar pixel size
  - Eiger 75µm vs 73µm
- MUCH faster
  - Eiger can collect images >200Hz
  - CCD detector ~20 images per minute with 1s exposure



Data Completeness

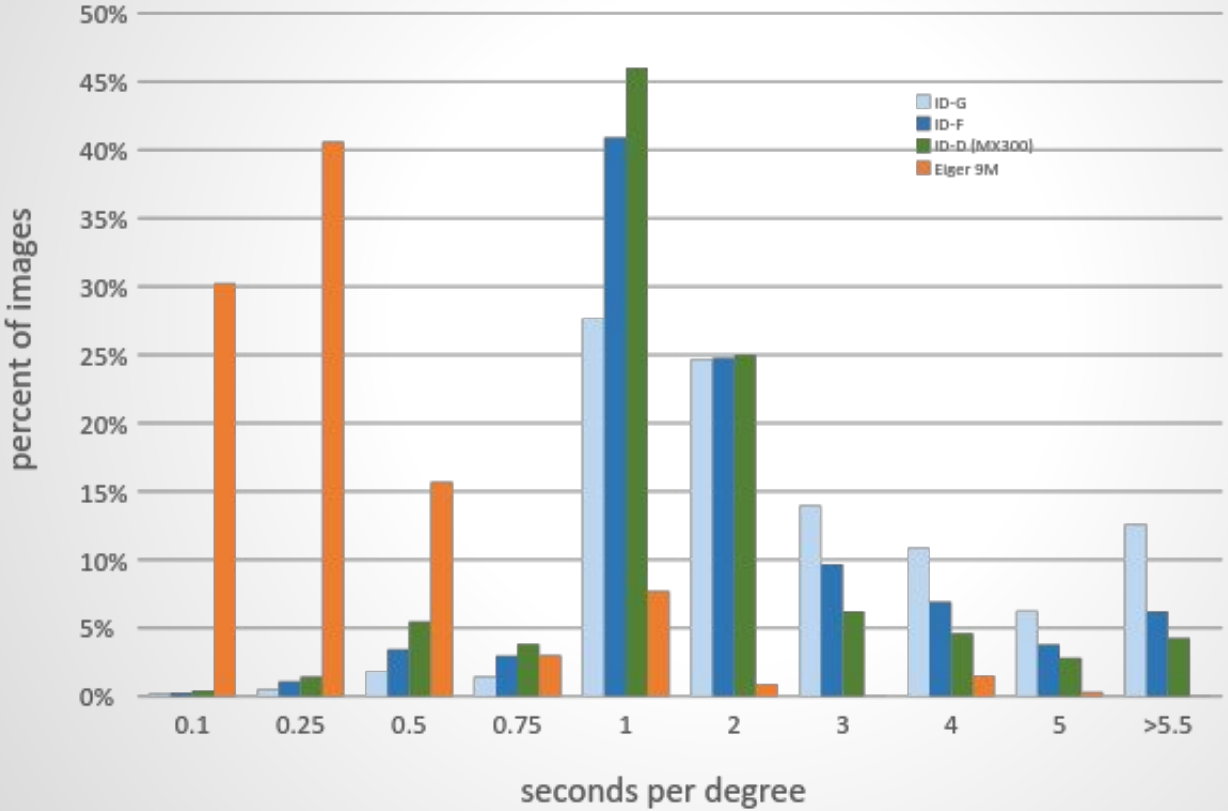




- 6% inactive area in gaps between modules
- Low point spread function
- Especially problematic with P1
- Location of gaps changes with distance/energy

Completeness and redundancy for a dataset processed as both P1 and P2. Only 200 of the 360 degrees were included in the monoclinic set to normalize for redundancy.

X-Ray Exposure



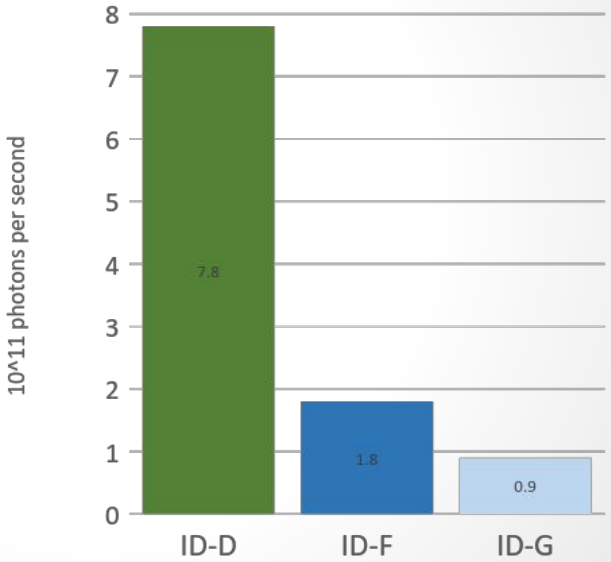
Shows how fast data is being collected

Does not consider:

- Attenuation
- Energy
- Beam focus
- Aperture size
- Slit size

Relative Beamline Flux at Sample

- Data collected by Elena Kondrashkina October 2015 using APS calibrated pin diode
- 50µm aperture ID-D
- 60µm aperture ID-F/ID-G
- 12.67 keV



Rotation Width

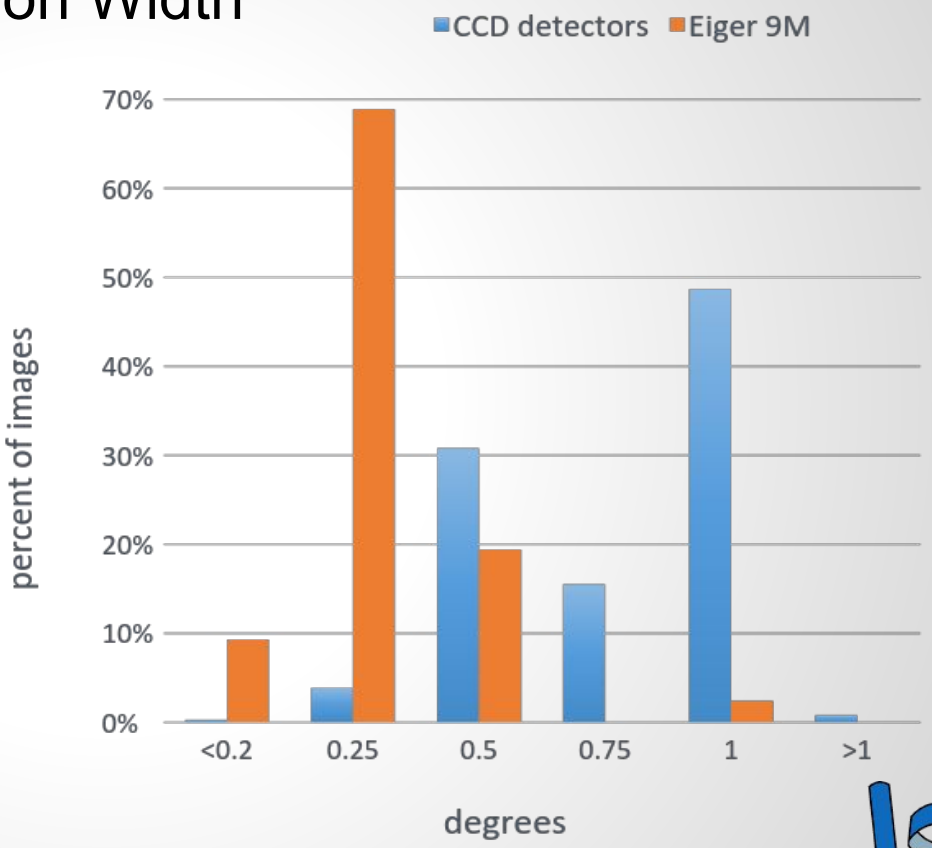
- how far crystal is rotated during exposure

- use continuous rotation with Eiger
  - 4μs gaps between frames

- optimum width dependent on mosaicity

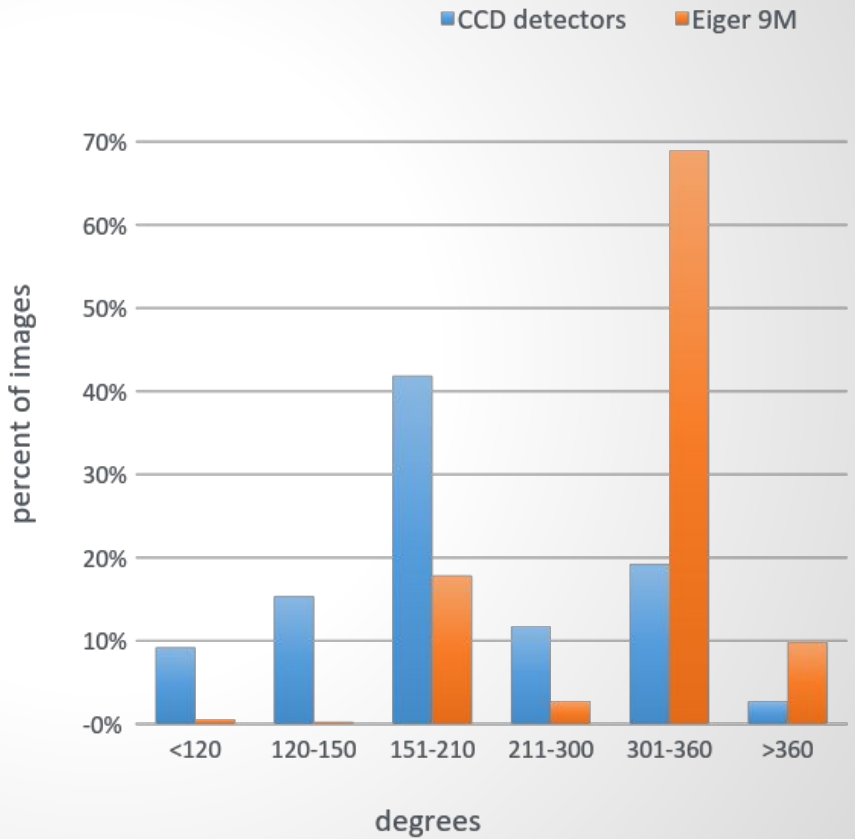
- CCD readout noise limits effectiveness
  - ½ to ¾ mosaicity on CCD
  - limited by collection time

- PAD's have no electronic or readout noise
  - 1/10 mosaicity proposed as target
  - limited by file size



Total Degrees Collected per Dataset

- collection on more samples
  - less screening before collection
  - delete useless sets after the fact
- collect 360 deg on everything
  - useful if twinned or wrong space group
  - collect past point of crystal dying
  - file format makes it difficult to delete frames
- very rarely use strategy
  - takes longer to run than collect dataset
  - exposing crystal more than necessary



Beamtime Allocation and Statistics

Beamtime Allocation Policies

1) The amount of beamtime to be allocated to a member is proportional to the dues that member is paying.

2) Members are able to, but are not required to, schedule time in advance of a run based on their allocation.

3) Members who have not scheduled their entire allocation may claim a rapid access slot at any time.

4) Members who have scheduled their entire allocation must wait one week prior to a rapid access slot before we will book that slot for them.

5) A scheduled slot is a right of first refusal. When an ESAF has not been submitted FIVE WORKING DAYS prior to your time then we may treat that a forfeiture of your right to use the beamtime. (The official APS policy is that ESAFs must be submitted one week prior to the start of beamtime.)

6) The general user program is subject to these same restrictions.

Allocated, Scheduled, and “Used” Beamtime Days from 2015-3 through 2016-2

	Shares	Allocated	Scheduled	“Used”
UM	1	61.0	78.8	68.2
MSU	1	61.0	34.8	14.0
VARI	1	61.0	64.5	27.7
WSU	0.25	15.3	16.3	15.4
NU	1	61.0	64.9	44.6
VUMC	0.5	30.5	36.8	27.2
UI	0.5	30.5	51.7	30.9
UW	0.5	30.5	31.2	19.8
Total	5.75	350.8	379.0	247.8

Note:

Most facilities, especially non-MX, report the Scheduled time as the Used time.

Here “Used” refers to the time between the first image and the last image and does not include any setup or other overhead.

LS-CAT Shares

	Original	2014	2015	2017
UM	1	1	1	1
MSU	1	1	1	1
VARI	1	1	1	0.25
WSU	1	0	0.25	0.25
NU	1	1	1	1
VUMC	0.5	0.5	0.5	0.5
UI	0.5	0.5	0.5	0.5
UW	0.5	0.5	0.5	0.5
Total	6.5	5.5	5.75	5.0